



NUTRITION

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- **Nutrition** may be defined as the **utilization of food by living organisms.**
- Food is the fuel source of the body.
- The ingested food undergoes metabolism to liberate energy required for the vital activities of the body.

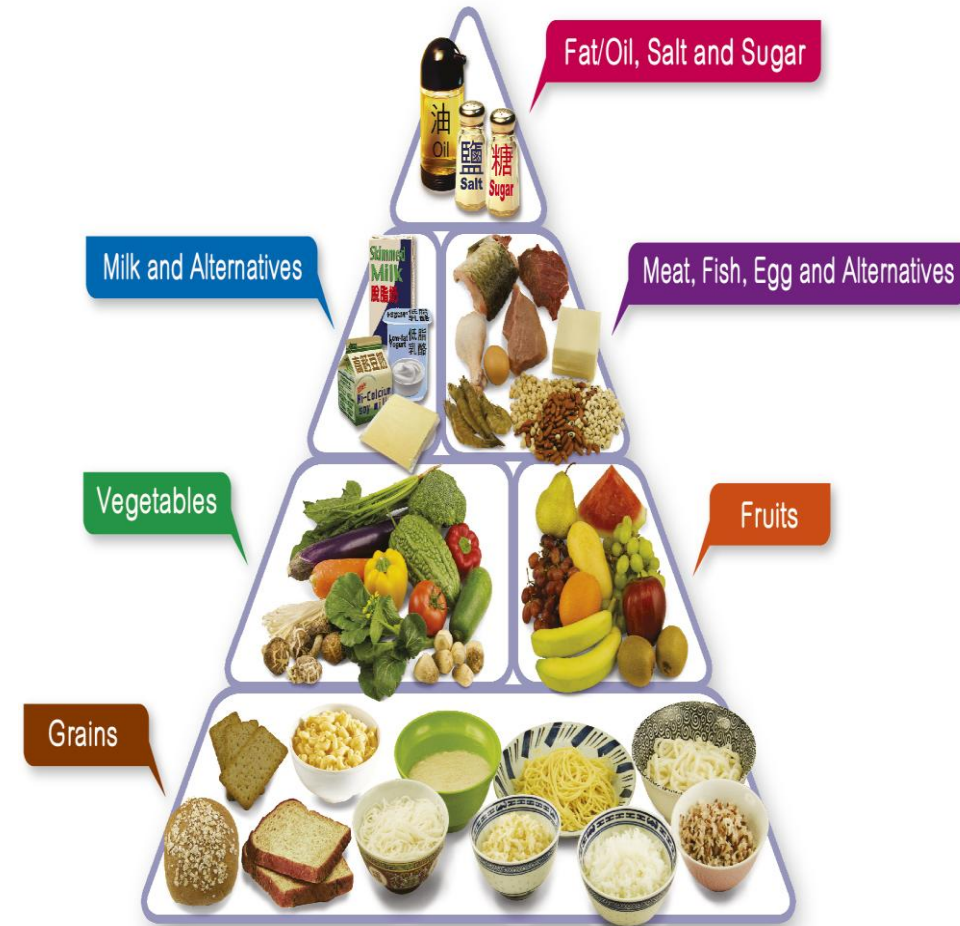


BALANCED DIET

- A *balanced diet* or *prudent diet* is defined as the diet which contains different types of foods, possessing the nutrients—carbohydrates, fats, proteins, vitamins and minerals—in a proportion to meet the requirements of the body.
- A balanced diet invariably supplies a little more of each nutrient than the minimum requirement to withstand the short duration of leanness and keep the body in a state of good health.



- The basic composition of balanced diet is highly variable, as it differs from country to country, depending on the availability of foods, social and cultural habits, besides the economic status, age, sex and physical activity of the individual largely influence the intake of diet.
- The Indian balanced diet is composed of cereals (rice, wheat, jowar), pulses, vegetables, roots and tubers, fruits, milk and milk products, fats and oils, sugar and groundnuts.
- Meat, fish and eggs are present in the non-vegetarian diets.
- In case of vegetarians, an additional intake of milk and pulses is recommended.



Balanced diet for an adult man

	Sedentary work		Moderate work		Heavy work	
	Vegetarian (g)	Non-vegetarian (g)	Vegetarian (g)	Non-vegetarian (g)	Vegetarian (g)	Non-vegetarian (g)
Cereals	400	400	475	475	650	650
Pulses	70	55	80	65	80	65
Green leafy vegetables	100	100	125	125	125	125
Other vegetables	75	75	75	75	100	100
Roots and tubers	75	75	100	100	100	100
Fruits	30	30	30	30	30	30
Milk	200	100	200	100	200	100
Fats and oils	35	40	40	40	50	50
Meat and fish	...	30	...	30	...	30
Eggs	...	30	...	30	...	30
Sugar and jaggery	30	30	40	40	55	55
Groundnuts	...	...	...	...	50	50



Balanced vegetarian diet for an adult male

	Sedentary	Moderate	Heavy
Cereals	290	340	450
Pulses	100	120	120
Leafy and other vegetables	200	200	200
Roots and tubers	200	225	250
Fruits	40	4040	
Milk	200	200	200
Fats and oils	35	3540	
Sugar and jaggery	35	3545	
Ground nut	10	1040	
Sprouts	50	5030	
All values in grams.			



CALORIFIC VALUES

- The calorific value (energy content) of a food is calculated from the heat released by the total combustion of food in a calorimeter.
- **Unit of heat : Calorie** is the unit of heat.
- One calorie represents the amount of **heat required to rise the temperature of one gram of water by 1°C** (i.e. from 15° to 16°C).
- A calorie is too small a unit. Therefore, it is more conveniently expressed as kilocalories (1,000 times calorie) which is represented by kcal or simply **Cal** (with capital 'C').



Calorific values of the three principal foodstuffs—carbohydrate, fat and protein

<i>Foodstuff</i>	<i>Energy value (Cal/g)</i>	
	<i>In bomb calorimeter</i>	<i>In the body</i>
Carbohydrate	4.1	4
Fat	9.4	9
Protein	5.4	4
Alcohol	7.1	7



- The carbohydrates and fats are completely oxidized (to CO_2 and H_2O) in the body; hence their fuel values, measured in the bomb calorimeter or in the body, are almost the same.
- Proteins, however, are not completely burnt in the body as they are converted to products such as urea, creatinine and ammonia, and excreted. Due to this reason, calorific value of protein in the body is less than that obtained in a bomb calorimeter.

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- **Alcohol** is a recent addition to the calorie (7 Cal/g) contribution, as it is a significant dietary component for some people.
 - It must be noted that the nutrients, namely vitamins and minerals, have no calorific value, although they are involved in several important body functions, including the generation of energy from carbohydrates, fats and proteins.

RECOMMENDED DIETARY ALLOWANCES (RDA)

- The recommended dietary/daily allowances (RDA) represents the *quantities of the nutrients to be provided in the diet daily* for maintaining good health and physical efficiency of the body.
- It must be remembered that RDA is not the minimum amount to just meet the body needs, but allowance is given for a safe margin.





Factors affecting RDA

1. **Sex** : The RDA for *men* is about **20% higher** than that for *women*. Iron is an exception as the requirement is greater in menstruating women. Additional requirements (20–30% above normal) are needed for pregnant and lactating women.
2. **Age** : In general, the nutrient requirement is much **higher in the growing age**. For instance, the protein requirement for a growing child is about 2 g/kg body wt/day compared to 1 g/kg body wt/day for adults.



Recommended dietary allowance (RDA) of important nutrients for an adult man, weighing 70 kg.

Nutrient(s)	RDA
Carbohydrates	400 g
Fats	70 g
Proteins	56 g*
Essential fatty acids	4 g
Vitamin A	1,000 µg **

Vitamin D	5 µg***
Vitamin E	10 µg
Vitamin K	70 µg
Ascorbic acid	60 mg
Thiamine	1.5 mg
Riboflavin	2 mg

Niacin	20 mg
Pyridoxine	2 mg
Folic acid	150 µg
Cobalamin	2 µg
Calcium	800 mg
Phosphorus	800 mg
Iron	10 mg



Recommended kcal allowances per day

ICMR recommendation		Infants	
Man (55 kg)		0–6 months	120/kg body wt
Sedentary work	2400	7–12 months	100/kg body wt
Moderate work	2800	Children	
Heavy work	4000	1–3 years	1200
Women (45 kg)		4–6 years	1500
Sedentary work	1900	7–9 years	1800
Moderate work	2200	10–12 years	2100
Heavy work	2600	13–15 years	
Later half of pregnancy	+300	Boys	2500
Lactation	+700	Girls	2300

 caloric requirements of adults per day (Cal/day)

Light work	—	2,200–2,500
Moderate work	—	2,500–2,900
Heavy work	—	2,900–3,500
Very heavy work	—	3,500–4,000



Recommended daily allowance (RDA) of proteins in different age groups

Nutrient	Requirement per day per kg body weight
Proteins	
Males	1 g
Females	1 g
Children	
Infants	2.4 g
Up to 10 years	1.75 g
Boys (> 10 years)	1.6 g
Girls (> 10 years)	1.4 g
Pregnancy and lactation	
Pregnancy	2 g
Lactation	2.5 g



BASAL METABOLIC RATE

- Basal metabolism or basal metabolic rate (BMR) is defined as the **minimum amount of energy required by the body** to maintain life at **complete physical and mental rest** in the postabsorptive state (i.e. 12 hours after the last meal).
- It may be noted that **resting metabolic rate(RMR)** is in recent use for BMR.



Factors affecting BMR

1. **Surface area** : The BMR is directly proportional to the surface area. Surface area is related to weight and height.
2. **Sex** : Men have marginally higher (about 5%) BMR than women. This is due to the higher proportion of lean muscle mass in men.
3. **Age** : In infants and growing children, with lean muscle mass, the BMR is higher. In adults, BMR decreases at the rate of about 2% per decade of life.
4. **Physical activity** : BMR is increased in persons (notably athletes) with regular exercise. This is mostly due to increase in body surface area.



5. **Hormones** : Thyroid hormones (T_3 and T_4) have a stimulatory effect on the metabolism of the body and, therefore, BMR. Thus, BMR is **raised in hyperthyroidism** and **reduced in hypothyroidism**.

The other hormones such as epinephrine, cortisol, growth hormone and sex hormones increase BMR.

6. **Environment** : In cold climates, the BMR is higher compared to warm climates.

7. **Starvation** : During the periods of starvation, the energy intake has an inverse relation with BMR, a decrease up to 50% has been reported. This may be an adaptation by the body.



8. **Fever** : Fever causes an increase in BMR. An elevation by more than 10% in BMR is observed for every 1°C rise in body temperature.

9. **Disease states** : BMR is elevated in various infections, leukemias, polycythemia, cardiac failure, hypertension etc. In Addison's disease (adrenal insufficiency), BMR is marginally lowered.

10. **Racial variations** : The BMR of Eskimos is much higher. The BMR of Oriental women living in USA is about 10% less than the average BMR of American women.



Normal values of BMR

- *For an adult man 35–38 Cal/sq. m/hr;*
- *for an adult woman 32–35 Cal/sq.m/hr.*
- A BMR value between –15% and +20% is considered as normal.
- Some authors continue to represent BMR as Cal/day.
- *For an adult man BMR is around 1,600 Cal/day, while for an adult woman around 1,400 Cal/day.*
- This is particularly important for easily calculating energy requirements per day.



Significance of BMR

- BMR is important to calculate the *calorie requirement* of an individual and *planning of diets*.
- Determination of BMR is useful for the assessment of thyroid function. In hypothyroidism, BMR is lowered (by about – 40%) while in hyperthyroidism it is elevated (by about + 70%).
- Starvation and certain disease conditions also influence BMR



Type of physical activity and energy expenditure (over and above BMR, about 65 Cal/hr).

Physical activity	Energy requirement (Cal/hr)
Sitting (quietly)	25
Standing (quietly)	30
Writing/eating/reading	30
Car driving	60
Typing	75
Household work (dish washing)	80
Walking (slow)	130
Sexual intercourse	140
Cycling (slow)	150
Running (moderate)	500
Swimming	600
Walking upstairs	800



For the sake of convenience, the individuals are grouped into four categories with regard to their physical activity and the requirement of energy.

Light work	—	30–40% of BMR(teachers, office workers, doctors)
Moderate work	—	40–50% of BMR(housewives, students)
Heavy work	—	50–60% of BMR(agricultural labourers, miners)
Very heavy work	—	60–100% of BMR(construction workers, rickshaw pullers)



SPECIFIC DYNAMIC ACTION

- The phenomenon of the *extra heat production by the body*, over and above the calculated caloric value, when a given *food is metabolized* by the body, is known as specific dynamic action (SDA).
- It is also known as *calorigenic action* or *thermogenic action* or *thermic action (effect) of food*.



SDA for different foods

- For a food containing 25 g of protein, the heat production from the caloric value is 100 Cal (25×4 Cal).
- However, when 25 g protein is utilized by the body, 130 Cal of heat is liberated.
- The extra 30 Cal is the SDA of protein. Likewise, consumption of 100 Cal of fat results in 113 Cal and 100 Cal of carbohydrate in 105 Cal, when metabolized in the body.
- SDA for protein, fat and carbohydrate are 30%, 13% and 5%, respectively.
- Thus, *proteins* possess the *highest SDA* while *carbohydrates* have the *lowest*.

SDA for mixed diet

- For a mixed diet, the SDA is not an additive value of different foods but it is much less.
- The presence of fats and carbohydrates reduces the SDA of proteins.
- Fats are most efficient in reducing SDA of foodstuffs.
- For a regularly consumed mixed diet, the SDA is *around 10%*.





Significance of SDA

- For the utilization of foods by the body, certain amount of energy is consumed from the body stores. This is actually an expenditure by the body for the utilization of foodstuffs.
- It is the highest for proteins (30%) and lowest for carbohydrates (5%) and for a mixed diet around 10%.
- It is, therefore, essential that an additional 10% calories should be added to the total energy needs (of the body) towards SDA, and the diet should be planned, accordingly.

- The higher SDA for protein indicates that it is not a good source of energy.
- Fat is the best source of energy due to its lowering effect on SDA.
- However, excessive utilization of fat leads to ketosis.

